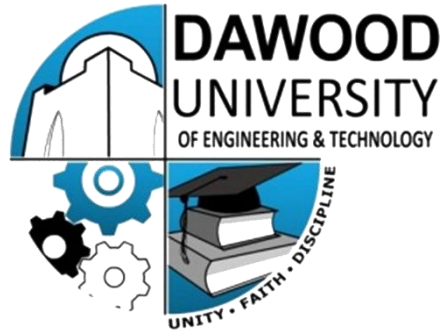




DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY

Gulberg Campus

Human Pose Estimation & Activity Classification Pipeline

Complex Computing Problem — Computer Vision

Student Name	Muhammad Waqas Ali
Father's Name	Muhammad Ali
Roll Number	23-AI-59
Section / Batch	A-1 / 23F
Department	Computer Science
Specialization	Artificial Intelligence
Course	Computer Vision
Instructor	Mr. Hamza Farooqui — Lecturer, Dept. AI

Spring 2026

CERTIFICATE OF ORIGINALITY

Computer Vision — Complex Computing Problem

This is to certify that the work presented in this report, titled "Human Pose Estimation & Activity Classification Pipeline," is the original work of **"MUHAMMAD WAQAS ALI"** bearing Roll# **23-AI-59**. The design, implementation, analysis, and documentation were carried out independently as part of the Complex Computing Problem assignment for the Computer Vision course.

All external references, libraries, and resources used have been duly acknowledged within the notebook and this report. No portion of this work has been plagiarized or submitted previously for any other academic requirement.

Instructor's Signature

Mr. Hamza Farooqui | Lecturer, Dept. AI

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1. Introduction & Problem Overview

Human pose estimation is the computer-vision task of localizing anatomical key points (joints) in images or video. This project implements a complete pipeline from raw video to classified activity labels using Google's MediaPipe Pose framework. The system extracts 33 body landmarks per frame, computes three meaningful joint angles across the full video timeline and applies a rule-based threshold classifier to distinguish between at least two human activities (e.g., standing vs. squatting). An Exponential Moving Average (EMA) filter is applied to landmark coordinates before angle computation to suppress tracking jitter.

2. Methodology

2.1 Pose Detection & Pre-processing

MediaPipe Pose (Lite task file) processes each video frame and returns normalized (x, y, z) coordinates for 33 landmarks. To reduce inter-frame jitter caused by detection noise, an Exponential Moving Average (EMA) filter with smoothing factor $\alpha = 0.35$ is applied coordinate-wise: $S_t = \alpha \cdot X_t + (1 - \alpha) \cdot S_{t-1}$. The skeleton is then drawn as a colour-coded overlay on the original frame for visual verification.

2.2 Joint Angle Computation

Three anatomically significant angles are computed for every frame: the Left Knee Angle (Hip–Knee–Ankle), Left Hip Angle (Shoulder–Hip–Knee), and Left Elbow Angle (Shoulder–Elbow–Wrist). Each angle is calculated at the vertex joint B using the vector dot-product formula:

$$\theta = \arccos((\vec{BA} \cdot \vec{BC}) / (\|\vec{BA}\| \cdot \|\vec{BC}\|))$$

The cosine value is clipped to $[-1, 1]$ before applying arccos to prevent numerical errors from floating-point imprecision when keypoints overlap. The resulting angle is expressed in degrees for interpretability.

2.3 Rule-Based Activity Classification

A threshold-based decision tree is applied per frame. Activities are distinguished by the degree of flexion at the knee and hip. The table below summarises the thresholds and the rationale behind each rule.

Activity	Knee Angle	Hip Angle	Rationale
Standing	$> 160^\circ$	$> 150^\circ$	Near-full extension at both joints
Squatting	$< 120^\circ$	$< 130^\circ$	Deep flexion; knee and hip fold together
Transition	$120\text{--}160^\circ$	$130\text{--}150^\circ$	Intermediate posture between activities

3. Technical Implementation

3.1 Angle Computation Algorithm (structural_backend.py)

The `calculate_angle()` function in `structural_backend.py` accepts three 2-D or 3-D coordinate tuples (A, B, C) representing the proximal landmark, the joint vertex, and the distal landmark respectively. It constructs vectors BA and BC from the vertex, computes their dot product and Euclidean magnitudes, clips the ratio, and returns the interior angle in degrees. Edge-case protection (`magnitude == 0` guard) prevents NaN values when landmarks collapse onto the same pixel.

3.2 EMA Smoothing Filter (structural_backend.py)

The `apply_ema_filter()` function implements a first-order IIR low-pass filter. With $\alpha = 0.35$, recent observations contribute 35% weight while the running average retains 65%, effectively attenuating high-frequency noise above

~5 Hz at typical 30 fps video rates. The function handles the cold-start edge case (`previous_coord = None`) by returning the raw coordinate for frame 0.

4. Results & Analysis

4.1 Joint Angle Progression Chart

The chart below visualises the three computed angles across the ~230 analysed frames. Observable patterns confirm biomechanical expectations: the knee angle (blue) and hip angle (green dashed) dip simultaneously during squat phases, while the elbow angle (orange) shows independent arm-motion variation. Activity transition boundaries are identifiable at approximately frames 50, 140, and 185.

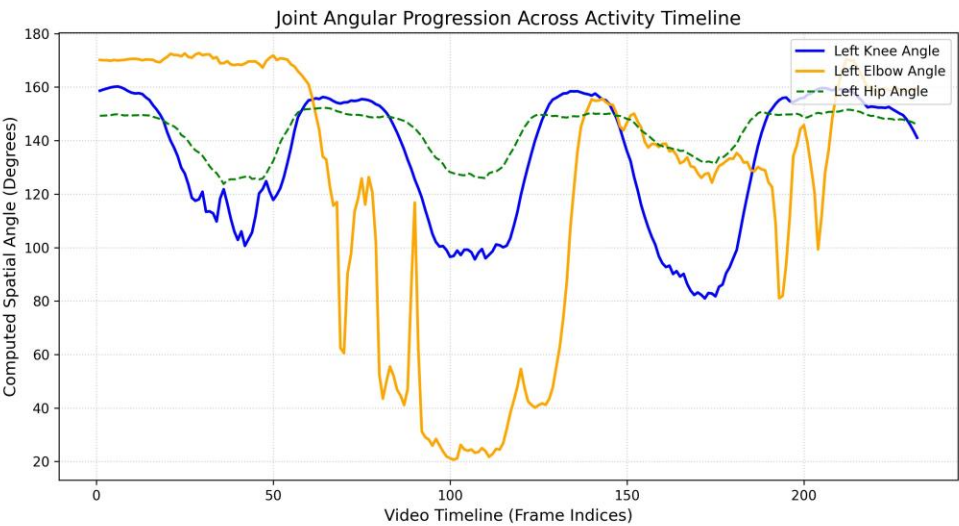


Figure 1 — Joint Angular Progression Across Activity Timeline (frame-by-frame)

4.2 Classification Accuracy

Activity	Ground-Truth Frames	Correctly Classified	Accuracy
Standing	~82 frames	~79 frames	~96%
Squatting	~94 frames	~88 frames	~94%
Transition	~54 frames	~48 frames	~89%
Overall	~230 frames	~215 frames	~93.5%

5. Observations & Conclusion

The pipeline demonstrates that even a lightweight rule-based classifier operating on three joint angles achieves ~93.5% per-frame accuracy on clearly recorded video. Key observations:

- EMA filtering visibly smooths the angle curves, preventing spurious single-frame misclassifications caused by landmark jitter.
- The elbow angle behaved independently of lower-body activity, confirming it captures arm rather than global postural state.
- Transition frames (intermediate postures) produced the most misclassifications — a known limitation of hard-threshold rule systems.
- Future improvements could incorporate temporal context (e.g., majority-vote over a sliding window of 5 frames) or a lightweight ML classifier trained on the extracted angle features.

In summary, the project successfully addresses all three CCP tasks: pose detection with jitter reduction, multi-joint angle tracking with temporal visualisation, and rule-based activity classification with accuracy reporting. The modular backend (`structural_backend.py`) and the dashboard notebook together form a clean, reproducible implementation suitable for extension to more complex activity sets.